Debaprashad Bhowmik

✆ 902-802-0129 | ➤ 1569 LeMarchant Street, Halifax, CA | ✉ *debaprashadbhowmik5314@gmail.com*
https://www.linkedin.com/in/debaprashad-bhowmik-74598a274/
Portfolio: *https://deb-portfolio-red.vercel.app/*

SUMMARY

Mechanical Engineering student with internship experience spanning predictive maintenance, digital twin modeling, HVAC/plumbing drafting, and technical documentation. Built physics-informed simulation workflows, trained machine learning models on engineering data, and supported mechanical design projects using Python, MATLAB, SolidWorks, AutoCAD, and Excel. Strong at turning technical analysis into clear drawings, reports, and decision-ready outputs.

EDUCATION

Bachelor of Mechanical Engineering September 2021– Present
Dalhousie University, Halifax, Canada

RELEVANT SKILLS

Technical Skills:

- **CAD, Drafting & Engineering Software:** Proficient in SolidWorks (part and assembly design), AutoCAD (2D drafting and 3D modeling) and MATLAB Simulink; foundational knowledge of REVIT and COMSOL.
- **Programming, Data Analysis & Machine Learning**: Python, MATLAB, C++, Excel, NumPy, Pandas, scikit-learn, Jupyter Notebook, Matplotlib, and Seaborn; experience in data analysis, machine learning model. development, and working with engineering and environmental datasets, including ERA5 and NOAA.
- **Simulation & Engineering Analysis:** Engineering simulation, interpretation of engineering drawings, technical note preparation, and analytical support for design problems.
- **Technical Writing & Documentation:** Technical reports, Statements of Work, engineering documentation, Word, and PowerPoint
- **Languages:** English, Bengali, and Hindi.

CAPSTONE DESIGN PROJECT

Bleeding Control Simulator September 2025 – April 2026
Dalhousie University | 6299 South St, Halifax, CA, NS B3H 4R2

- Collaborated in a 4-member capstone team to design and prototype a portable bleeding control simulator for BraveHeart First Aid, improving realism in first-aid training for direct pressure, wound packing, and tourniquet interventions.
- Engineered an integrated system comprising a medical prosthetic, fluid reservoir, pump, compact tubing network, and recirculating flow loop to simulate blood loss up to 800 mL and bleeding rates up to 70 mL/min.
- Helped developing the electronic feedback and control system by integrating pressure sensing, volumetric sensing, Arduino-based control logic, and a MATLAB-based interface, enabling LED and on-screen feedback during simulation events.
- Supported pump and tubing selection through fluid-flow analysis and prototype testing to satisfy target operating conditions of 660 Pa to 18 kPa, with early validation testing recording 1536 mL/min volumetric flow.
- Optimized a low-cost fake blood formulation through iterative recipe evaluation and viscosity-focused testing, selecting a water, vegetable oil, and food colouring mixture costing approximately $0.51/L, well below the $30/L design limit.
- Prepared technical documentation and client hand-off material covering setup, operation, cleaning mode, maintenance, and future modification, supporting delivery of the simulator within the project's $1,850 budget.

WORK EXPERIENCE

Marine Engineering Intern January 2025 – May 2025
BMT Canada | 780 Windmill Rd, Dartmouth, CA, NS B3B 1T3

- Developed a physics-informed, environment-aware digital twin of Caterpillar C27/C32 ACERT V-12 marine diesel engines to support predictive-maintenance research for Royal Canadian Navy applications.

- Generated a two-year, hourly synthetic dataset with 17,520 records across 34 channels by combining engine physics with ERA5 and NOAA environmental data for Arctic, mid-latitude, and tropical operating conditions.
- Trained and evaluated machine-learning models on simulated engine data to support anomaly detection and predictive maintenance, using Python (scikit-learn and Jupyter) for regression/classification and performance evaluation.
- Modelled degradation in 10 critical components and injected 6 fault modes using Weibull-informed wear laws, Poisson-triggered events, and sensor noise/drift logic, yielding 142 labelled fault events for downstream model development.
- Validated simulator outputs against OEM sea-trial data and published MTBF benchmarks; healthy-operation KPIs remained within stated acceptable error bands, and anomaly-testing performance reached ROC-AUC 0.93 and F1-score 0.81 with drift compensation.
- Authored a formal technical report and supporting visualizations documenting model architecture, governing equations, fault logic, validation results, and recommendations for predictive-maintenance deployment.

Mechanical Engineering Intern May 2024 – August 2024

Trimac Engineering | 45 Wabana Ct, Sydney, NS B1P 0B9

- Created detailed HVAC and plumbing drawings in AutoCAD, including boiler schematics and system layouts, under supervisor guidance, while gaining a strong understanding of mechanical drawings and symbols.
- Collaborated with engineers from multiple disciplines to select and reposition a centrifugal electric pump based on client requirements, updating the existing drawing accordingly using AutoCAD.
- Conducted site visits to a power plant, observing turbines, boilers, and other systems in operation, gaining hands-on experience and further supporting theoretical knowledge of turbines.
- Developed and documented Statements of Work (SOW) using MS Word, ensuring proper indentation and professional writing, showcasing technical writing skills and proficiency in MS Word.
- Created and formatted detailed mechanical notes and annotations in AutoCAD, showcasing technical writing skills by ensuring proper formatting, organization, and attention to detail for accurate and compliant instructions to the contractor regarding HVAC and plumbing systems.
- Participated in a site visit to a hotel's mechanical room, performing markups for the pump, boiler, and associated pipe connections, which were later translated into AutoCAD drawings.
- Solved a cantilever beam force problem that was implemented in a real-world project.
- Designed a coupling bolt for a hydroelectric power station, which was fabricated and used on-site.
- Prepared technical reports and supporting documentation based on engineering tasks, site visits, and project findings, ensuring clear and professional communication of technical information.

Mechanical Engineering Intern September 2023 – December 2023

Integrated Project Services – IPS | Toronto, Ontario, Canada

- Gained hands-on experience with Revu Bluebeam software, leveraging it to review and analyze GMP (Good Manufacturing Practice) facility floor plans, enhancing comprehension of project documentation and compliance standards.
- Contributed to the design process of HVAC systems for a GMP facility, authoring a comprehensive technical memorandum that demonstrated a clear understanding of design principles, regulatory requirements, and system integration.
- Managed and updated Excel datasheets, ensuring meticulous attention to detail in data accuracy, validation, and analysis, aiding project tracking and reporting.
- Developed foundational knowledge of REVIT software, acquiring skills in mechanical design and 3D modeling.
- Collaborated closely with a senior mechanical designer, gaining insight into mechanical design methodologies and the practical application of engineering concepts in real-world projects.

VOLUNTEER EXPERIENCE

General Crew Member (Thermal Subsystem) September 2023 – January 2024

Dalhousie Space Society - Space Division | Dalhousie University | 6299 South St, Halifax, CA, NS B3H 4R2

- Actively participated in the MENTIS CubeSat program within the thermal team, with the objective of gathering crucial information for the Canadian Space Agency (CSA).
- Utilized COMSOL and MATLAB for thermal simulation, analyzing temperature dynamics of the CubeSat.
- Employed SolidWorks to create a detailed virtual twin of the CubeSat, aiding in design validation and optimization.